# Printune: how the app flows

Kalico plugin + Mainsail page. Every test is a print (or a measurement) followed by a pick; results go into Orca presets, machine values into printer.cfg. State: branch initial-import, 2026-10-15.

### PRINTUNE macro (Mainsail console)

- The only visible macro. Opens a button menu with every workflow below; each step then talks to you through Mainsail prompts.
- Greyed buttons when a tool is missing (no accelerometer, not CoreXY, no Shake&Tune); tapping one links to the tool's docs.

### Printune page (Mainsail sidebar)

- Same workflows with more options: setup form, live progress, big pick buttons, babystep buttons, graphs, preset download. Made for phones down to 320 px wide.
- Picks can be made on the page or with the console prompts.

## 0. Pre-flight

Once per printer, before any filament

### 1 Machine setup (required before any test print)

- Bed area: defaults to the bed mesh area (mesh_min / mesh_max) cut to the axis limits; typed in when there is no bed mesh.
- Measured by you: toolhead box (−X, +X, −Y, +Y from the nozzle), bed to X beam height at Z=0, and nozzle protrusion below the rest of the toolhead.
- Risk notice: Printune and its authors take zero responsibility; you confirm you measured the values. Changing any value clears the acceptance.
- Until done, every printing test is locked (wizard, single tests, Advanced, Orca and command line). Session files carry the geometry and are refused if it no longer matches.

printer.cfg [printune]: bed_area, toolhead_box, gantry_height, nozzle_protrusion, setup_accepted

### 2 Belt tension (accelerometer pluck)

- Needs CoreXY + an accelerometer on the toolhead ([resonance_tester]); otherwise greyed out.
- Set the target frequency from your build guide (±1 / ±5 Hz) and 3 or 5 plucks. The toolhead parks in the middle so the belt span is known.
- Hands off for 1.5 s: fans and motors are recorded and subtracted later.
- Pluck belt A like a guitar string, about once a second. Printune finds each pluck, takes its frequency (checks for a tone at half the strongest peak) and shows the average against the target.
- Then belt B. CoreXY belts pull on each other, so both are measured every round.

Shows: A 104 Hz (avg of 5), 6 Hz low · nothing stored; raw recordings saved in printune_exports/recordings for download

### 3 Turn advice, repeat

- First rough: both belts to about 90-100 Hz.
- Then small turns: 'Tighten belt B about 3/8 turn, then measure both again'. It learns Hz per turn and how much one belt drags the other from each turn you report.
- Done when both are within belt_tolerance (2 Hz) of the target and of each other.

Advice only

*Loop: measure A, measure B, turn, until both are in tolerance.*

### 4 Shake&Tune belt graph (optional)

- Offered when the belts are done, if Shake&Tune is installed: runs its belt comparison and shows the graph under the prompt and on the page.
- Not installed: greyed button linking to the Shake&Tune project.

Graph only

### 5 Z offset (starting value)

- The printer's own paper test: PROBE_CALIBRATE with a probe, BEACON_CALIBRATE with a Beacon, Z_ENDSTOP_CALIBRATE without a probe.
- Printune homes, levels the gantry (Z tilt / QGL) if it can and moves to the middle of the bed. Lower the nozzle until the paper drags, Accept, Save config.
- Redo after a nozzle or build plate swap. The first layer test fine-tunes it while printing.

printer.cfg: probe z_offset / position_endstop (SAVE_CONFIG)

### Hands-on tools (no print, any time)

#### Extruder calibration

- Mark the filament 120 mm above the extruder, extrude 100 mm, type what is left. Printune computes the new rotation_distance.

printer.cfg: rotation_distance (stays fixed; per-filament differences go in Flow)

#### Extrude-in-air flow check

- Extrudes 100 mm at a chosen mm³/s; raise it until less than 100 mm goes in. A quick check next to the printed max flow tower.

## 1. Standard profile tuning

Tuning Wizard (big blue button), per filament

### 1 Setup

- Pick the filament type (PLA, PETG, ABS, ASA, PC, PA, TPU...), then the nozzle range from the box and the bed temperature. Defaults come from the type.
- Preset name: '<type> Printune' (rename on the page). Heat soak on/off is only asked when the printer is marked enclosed.
- Run the whole wizard, or a single test from the same list.

### 2 Heat soak (enclosed printers only)

- Extruder heater off, toolhead parked as far from the bed as possible; the nozzle thermistor reads the chamber. Ends when that reading levels off (between soak_min and soak_max), or Skip.
- The printer stays hot between all tests that follow.

### 3 First layer squish

- Prints 9 patches of 25×25 mm over the whole bed. Babystep with ±0.01 / ±0.05 mm while they print.
- 'Squish is good', or 'Print 4 more patches' to check the new offset.

Fine-tunes the pre-flight Z offset (machine setting): Z_OFFSET_APPLY_PROBE + SAVE_CONFIG; never in a filament preset

### 4 Pressure advance

- A row of hollow frames, one per value (0-0.08 in 0.005 steps). Each edge prints slow-fast-slow, so a wrong value shows a bulge or gap where the speed changes.
- Pick the most even frame; optional fine pass in 0.001 steps.

Filament › Pressure advance (Enable ticked)

### 5 Flow (extrusion multiplier)

- 30×30×3 mm cubes at −4 ... +4 % flow, numbered, printed with the picked PA.
- Pick the smoothest top surface; optional fine pass in 0.5 % steps.

Filament › Flow ratio

### 6 Retraction and temperature

- Up to 4 towers across the temperature range, hottest first, each with rings at 0-1.0 mm retraction.
- Pick the cleanest tower (temperature), then the ring 1-2 above where stringing stops. Skippable: direct drive often needs no tuning.

Filament › Nozzle temperature + retraction (SET_RETRACTION in filament start G-code, or retraction overrides)

*On small beds the tests spread over several plates; Printune pauses with 'clear the plate'.*

### 7 Results and preset

- Summary of every pick. Download the Orca filament preset (JSON) from the page and import it.
- Values are applied live only during the session; the printer's own values come back afterwards.

printune_exports/<name>.json → Orca filament preset (single source of truth)

### 8 Input shaper (last)

- Needs an accelerometer and [input_shaper]; otherwise greyed (link: Kalico 'Measuring Resonances').
- Runs Kalico's SHAPER_CALIBRATE: shows the pick per axis and the highest acceleration that keeps corners sharp. On right away; 'Save to printer.cfg' runs SAVE_CONFIG.
- With Shake&Tune: its axis graphs run from the same step.
- If the shaper changed: offers a pressure advance re-check.

printer.cfg: shaper_type / shaper_freq per axis

## 2. Advanced tuning

Per filament, after the wizard; same preset

### 1 Setup (same as the wizard)

- Use the wizard's preset name: Advanced prints with the wizard's temperature, PA, flow and retraction, and adds its results to that preset (fields it did not test are kept). A different filament type gets a new preset instead.
- Heat soak again if enclosed. Each step is skippable and can also run alone.

### 2a Cooling: fan %

- Overhang blocks, one per fan speed (PLA/PETG 40/70/100 %, ABS/ASA 20-80 %, PC/PA from 0 %), with 30° to 70° overhang bands. The outer wall prints at its normal speed; layers are padded to a typical layer time.
- Pick the lowest fan % whose overhangs look good. One constant fan (varying fan causes banding).

Filament › Cooling: fan min = fan max, overhang fan, keep fan always on

### 2b Cooling: minimum layer time

- A cone printed on its own at that fan %, constant speed, no slowdown: layer time falls with height (ABS 26 s, PLA 16 s down to 1 s), marker every 2 s.
- Pick the marker where the tip starts to melt, sag or curl. Printune adds 3 s; for ABS it flags results far from about 15 s. Curling at low fan can also be bed adhesion.

Filament › Cooling: slow down layer time (Orca's minimum layer time)

*Runs after the soak, so it matches your chamber temperature and part cooling.*

### 3 Bridges

- Strips bridging 20, 40 and 60 mm between 4 mm pillars, in three passes, each building on the pick before it.
- Fan: from the cooling pick up to 100 % (skipped if that is already 100 %). Speed: 0.6 to 2 × the bridge speed. Flow: 0.85 to 1.00.
- Pick the cleanest, flattest strip in each pass.

Filament › Overhang/bridge fan speed

Process › Bridge speed, bridge flow ratio (a small process preset to import)

### 4 Max volumetric flow tower

- A 60 mm round single-wall spiral. Flow rises in 3 mm bands (default 5-30 mm³/s, TPU 1-12); a small bump marks every 5 mm³/s. Line 1.5 × nozzle wide, 0.75 × nozzle tall.
- Nothing slows it: corners, speed limits and acceleration are checked; bands the printer can't reach are dropped with a warning.
- Pick the band (or type a height) where the wall gets rough or weak. Printune proposes 90 % of it and shows the matching mm/s at 0.2 × 0.45 mm.

Filament › Max volumetric speed

*Method as Orca's own: flow = start + height × step, minus a 10-20 % margin. Printed last, alone on its own plate: at 79 mm it is far taller than the gantry clearance.*

### 5 Max speed and acceleration (machine) — PLANNED

- Fast test moves, then re-home and compare positions: more than one full step of drift means it skipped. Keep about 15 % below the limit found.
- Only proves the flow on the test printer (it can't lose steps).

printer.cfg: max_velocity / max_accel

Orca printer profile: limits

### ? Shrinkage — PLANNED

- Measure a printed cube with calipers (ABS/ASA/PC/PA print undersized).

Filament › Shrinkage (XY, Z)

*Suggested, not yet confirmed.*

## 3. Expert mode

Your own workflows (planned)

### 1 Build a workflow — PLANNED

- A palette of every Printune module (pre-flight, standard, advanced, custom). Drag them into order (up/down buttons too, for phones).
- Each step keeps its own settings: ranges, step sizes, skip if already tuned.
- Order warnings, not blocks (e.g. 'flow before PA gives a worse pick').

### 2 Custom test print — PLANNED

- Upload G-code you sliced yourself, e.g. a 200 % Voron cube.
- Describe the tower in Kalico TUNING_TOWER terms: COMMAND (e.g. SET_PRESSURE_ADVANCE), PARAMETER (ADVANCE), START, then FACTOR or STEP_DELTA + STEP_HEIGHT, optional BAND.
- Pick where the result goes: an Orca field or nowhere.
- After the print, type the best height or band. Printune turns it back into the value and stores it.

Any chosen Orca field, e.g. Filament › Pressure advance

### 3 Manual step — PLANNED

- Instructions plus a value box, for anything Printune can't print or measure itself.

### 4 Save and share — PLANNED

- Workflows are files in printune_exports/workflows. The standard wizard is the built-in workflow: copy it and edit the copy.

## Where results go

- Orca filament preset (per filament)
- Orca process / printer profile
- printer.cfg (machine values)
- Advice or graph only, nothing stored
- Dashed box = planned, not built yet
- 3 Order within a workflow